

Lecture 4 (9/21)

Continuous Distributions

The uniform distribution

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In code

```
y = np.random.uniform(a, b, size=1000)
```

Example

The time I wait at the bus stop in minutes is $T \sim \text{Uniform}(0, 5)$ (not very realistic).

- What is the probability I wait more than 2 minutes?
- What is the average time I wait, given that I wait longer than 3 minutes?

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Example

Consider

$$\begin{aligned}X &\sim \text{Bernoulli}(1/4) \\ Y|X &\sim \text{Uniform}(0, 1 + X)\end{aligned}$$

- Compute (i) $E[Y]$ and (ii) $P(Y = 1|X > 1/2)$.
- Translate this problem into code and estimate these from samples.

Example

We often use $\text{Uniform}(0, 1)$ to model an unknown probability. Say Y is the outcome of a binary election. Before we do any polling we have complete ignorance about the result. We model this as

$$\begin{aligned} X &\sim \text{Uniform}(0, 1) \\ Y|X &\sim \text{Bernoulli}(X) \end{aligned}$$

Densities

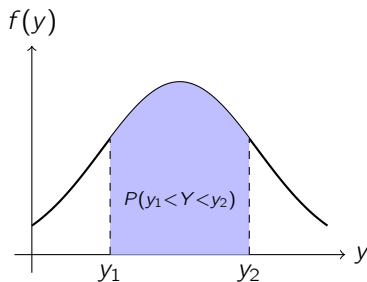
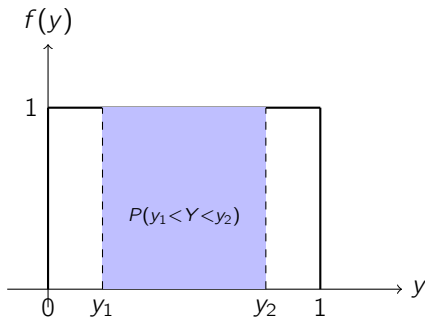
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Densities

A continuous random variable Y has a density $f(y) \geq 0$ with

$$P(a < Y < b) = \int_a^b f(y) dy, \quad \int_{-\infty}^{\infty} f(y) dy = 1.$$

For the uniform case, $f(y) = \frac{1}{b-a}$ on $[a, b]$. Note $f(y)$ can exceed 1; the only requirement is that the total area under f equals 1.

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- Often we use continuous random variables to model things like grades, or the number of people, because from a mathematical perspective they can be simpler to work with.
- The most important thing for this class is to understand conceptually how probability calculations work, not to evaluate the integrals to get exact results.

Density vs. CDF

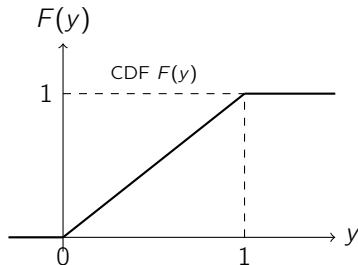
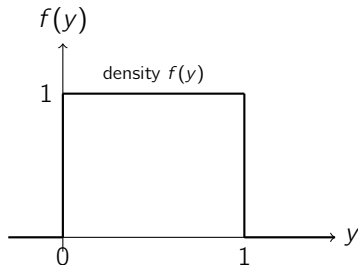
- The density $f(y)$ tells us how probability is spread out: probabilities come from *areas* under f .
- The CDF (cumulative distribution function) accumulates that area from the left:

$$F(y) = P(Y \leq y) = \int_{-\infty}^y f(t) dt \implies f(y) = F'(y)$$

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Example

Let $U \sim \text{Uniform}(0, 1)$ and find the CDF and densities of U^2 and $\sqrt{U}$.

Example

Let $U \sim \text{Uniform}(0, 1)$ and define $T = -\frac{1}{\lambda} \log(U)$. Find the CDF and density of T .

Exponential distribution: a constant-rate process

The exponential distribution is the natural model for the waiting time until an event that occurs at a **constant rate** λ – e.g. a bus arriving, a customer walking in, a radioactive atom decaying, a machine part failing.

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- “Constant rate” means: in a short interval dt , the chance of the event happening *in the next instant* is the same λdt , no matter how long we’ve already waited.
- In symbols, if T is the waiting time,

$$P(t < T \leq t + dt \mid T > t) \approx \lambda dt \quad \text{for every } t.$$

- This “no memory of how long you’ve waited” property is called **memorylessness**:

$$P(T > t + s \mid T > t) = P(T > s).$$

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Because the rate never changes, $E[T] = 1/\lambda$: on average you wait one “rate unit” of time for the event.

Solutions

Bus stop, $T \sim \text{Uniform}(0, 5)$:

- $P(T > 2) = 3/5 = 0.6$
- $E[T \mid T > 3] = 4$ (since $T \mid T > 3 \sim \text{Uniform}(3, 5)$)

$X \sim \text{Bernoulli}(1/4)$, $Y \mid X \sim \text{Uniform}(0, 1 + X)$:

- $E[Y] = 5/8 = 0.625$
- $P(Y = 1 \mid X > 1/2) = 0$ (continuous, so $P(Y=y) = 0$ for any y)

$U \sim \text{Uniform}(0, 1)$:

- U^2 : CDF = $\sqrt{v}$, density = $\frac{1}{2\sqrt{v}}$ on $(0, 1)$
- $\sqrt{U}$: CDF = w^2 , density = $2w$ on $(0, 1)$
- $T = -\log(U)/\lambda$: CDF = $1 - e^{-\lambda t}$, density = $\lambda e^{-\lambda t}$, i.e. $T \sim \text{Exponential}(\lambda)$